Project Presentation

Sharp Wave Ripples in a Bistable Network

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Structure

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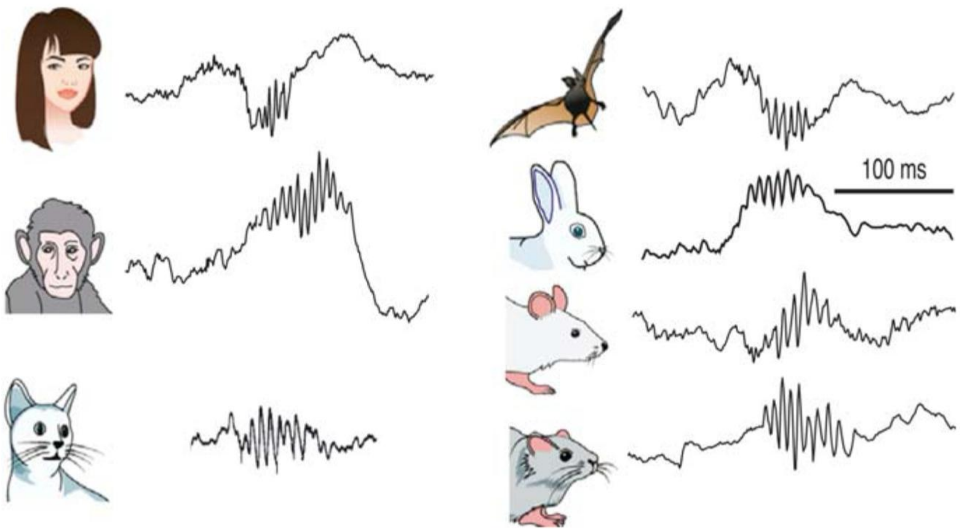
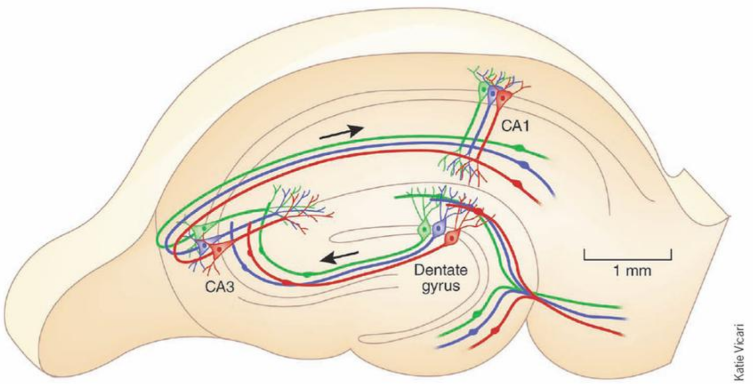


FIGURE 2. Preservation of SPW-Rs in mammals. Illustrative traces of ripples recorded from various species. Reproduced from Buzsáki et al., (2013).

Hippocampus

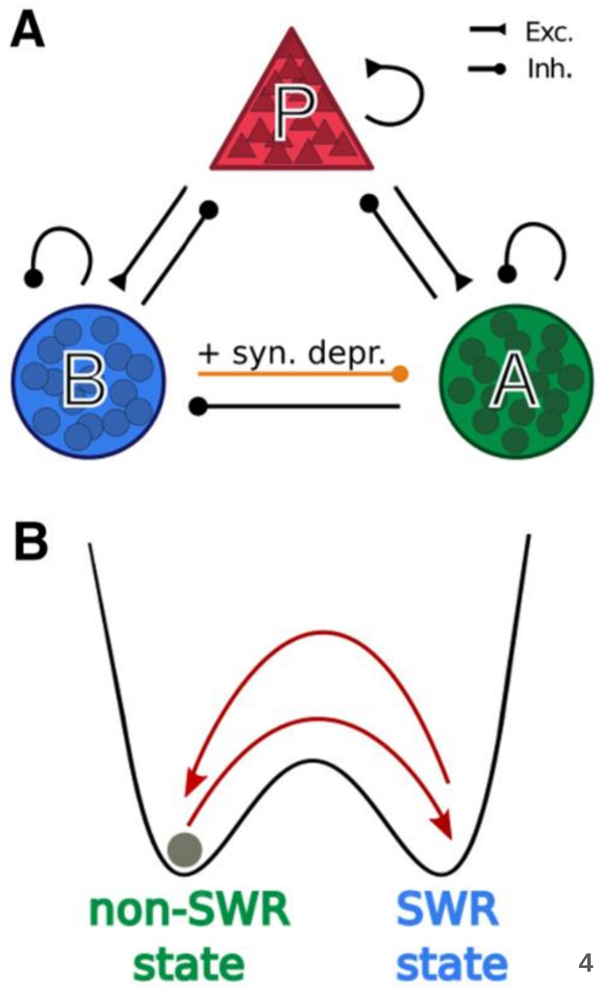
Introduction: Biological Background

- Sharp Wave Ripple (**SWR**): 50-100 ms
- elevated, synchronized network activity
- originating in CA3
- cells participating: Pyramidal Cells (**P**), PV+ Basket Cells (**B**)
- occurs during slow-wave sleep, awake rest
- involved in memory consolidation



Introduction: Model Mechanism

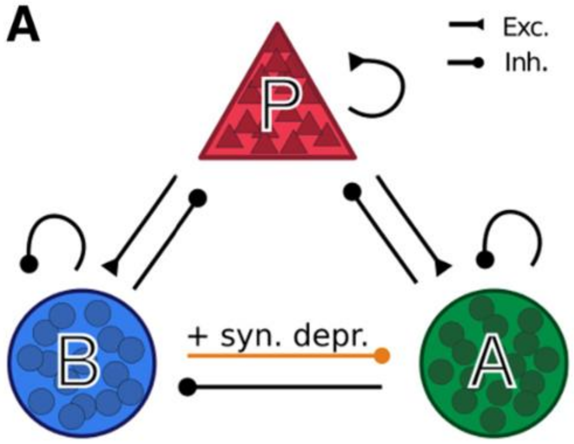
- Network of three neuronal populations:
  - P, B, A (**A**nti-SWR cells)
- New cell population “A” proposed due to findings by Schlingloff et al. (2014)
- Induction of SWR events by stimulation of basket cells (B)
- potentially due to **disinhibition mechanism**



Introduction: Network Dynamics

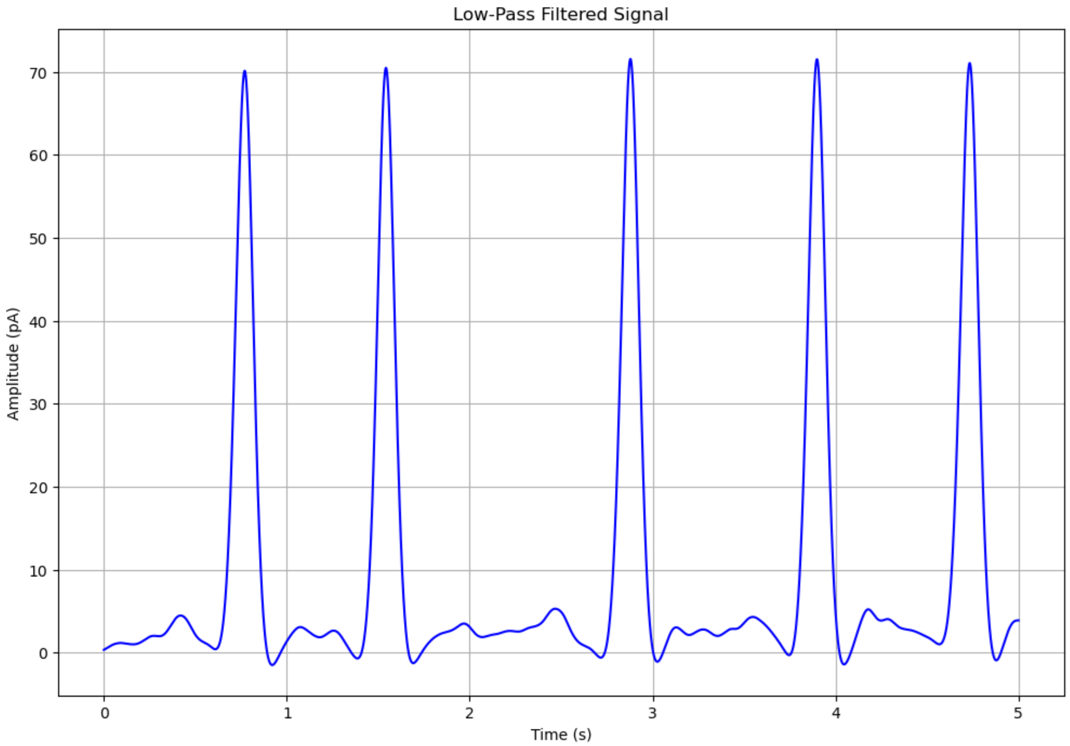
- Disinhibition modelled through population A
- A inhibits B and P in non-SWR state
- Synaptic efficacy B→A increases over time, inhibition of A increases
- inhibition of P and B is released → SWR state
- short-term synaptic depression mechanism leads to “depletion” of B to A synapses → non-SWR state

→ **bistable network**

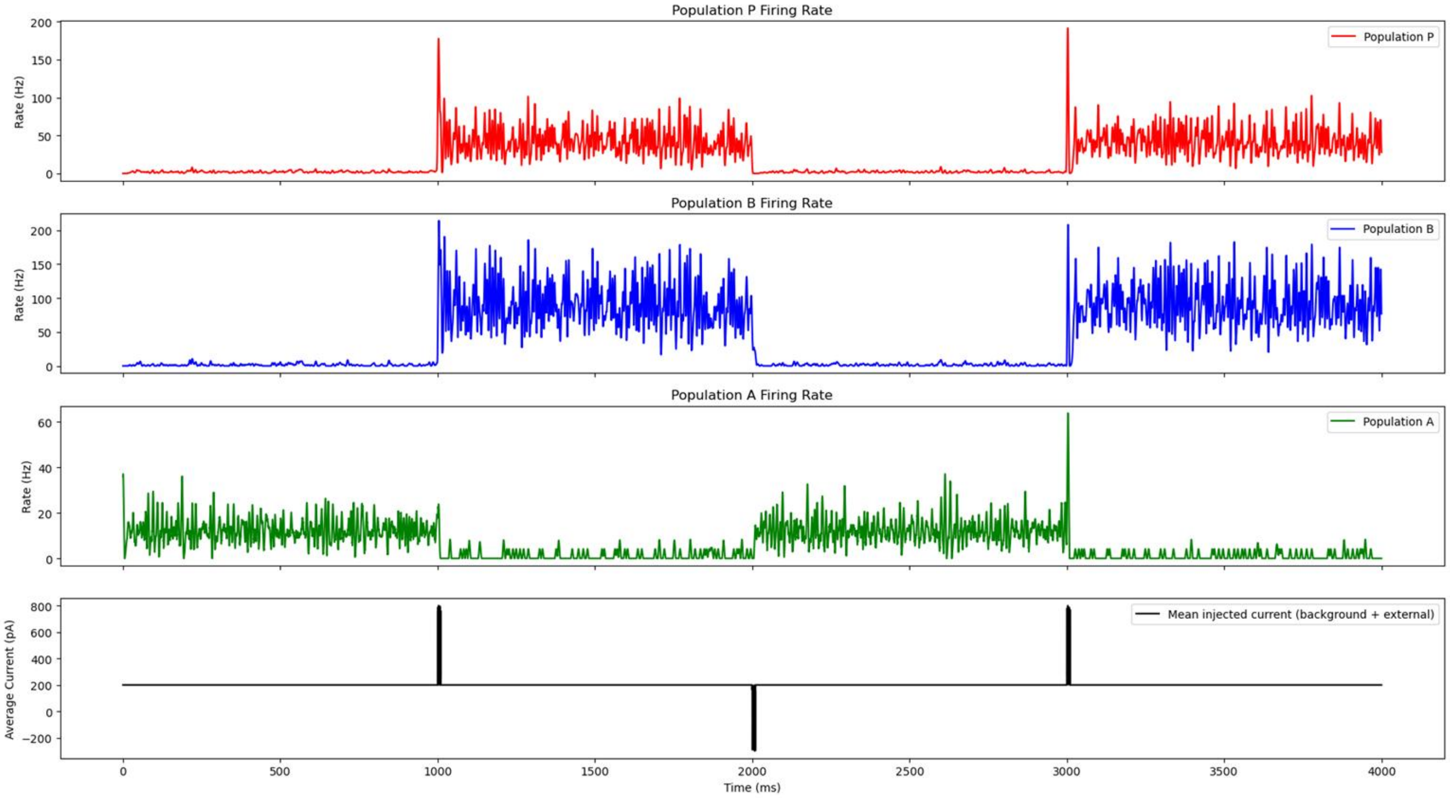


Methods

- LIF neuron model with conductance-based synapse
- sparse connectivity between populations
- LFP was estimated for comparability to extracellular recordings
- statistical analyses

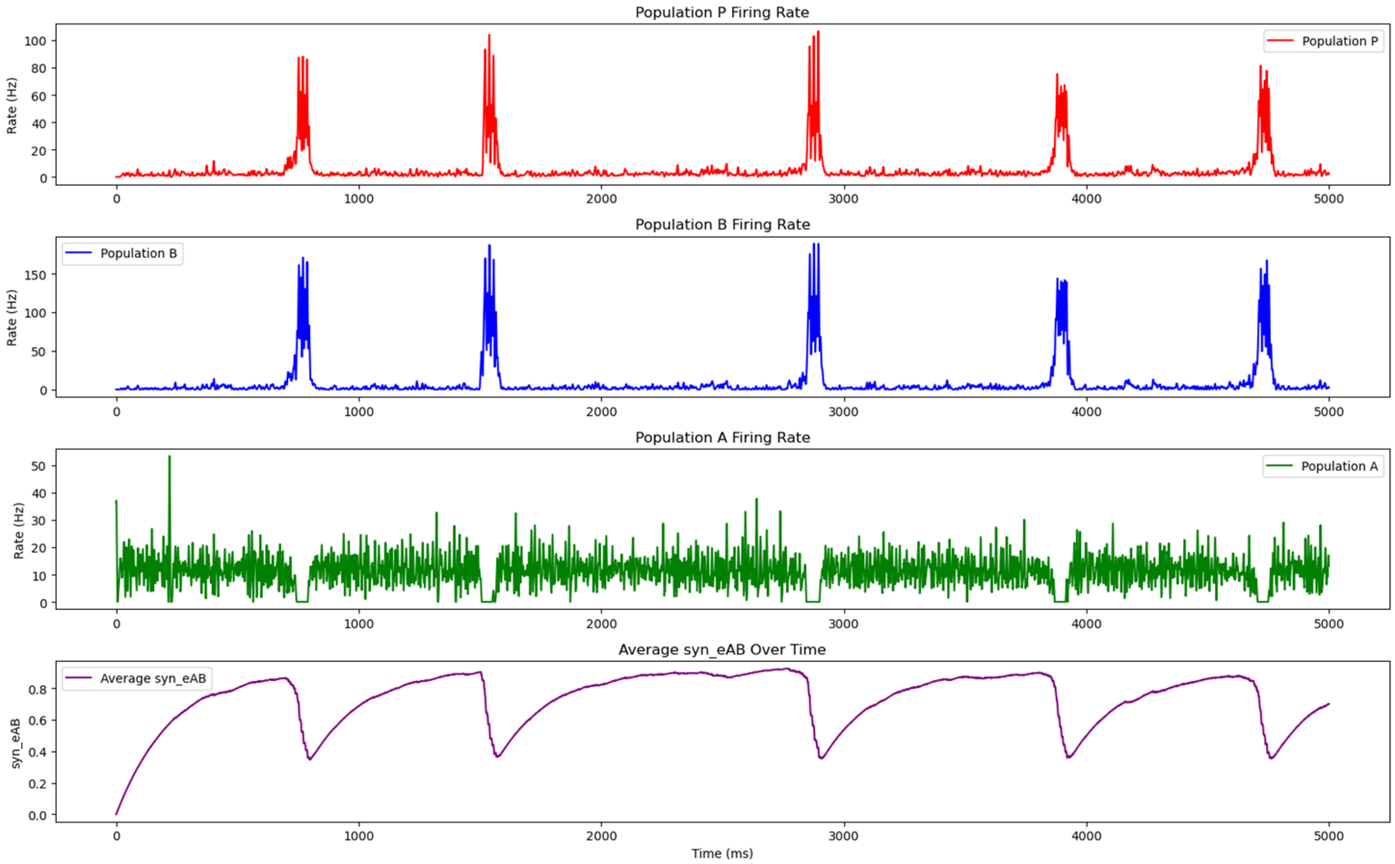


Results: Bistability



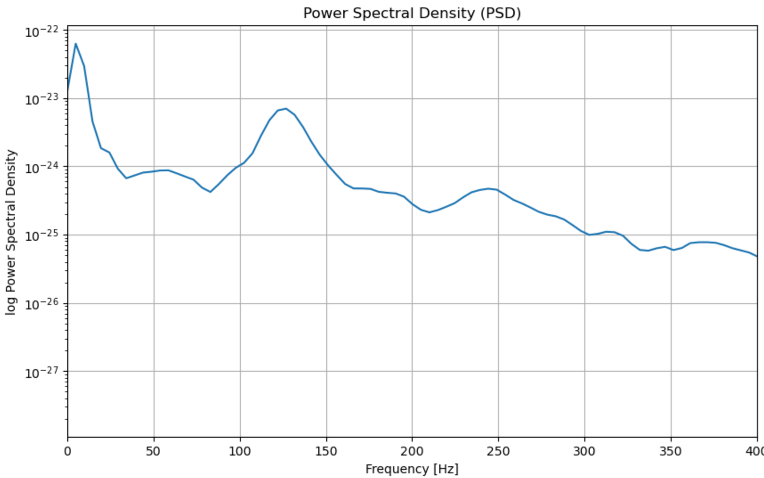
- synaptic efficacy (B -> A) clamped
- 10 ms current pulses to B population

Results: Spontaneous SWR

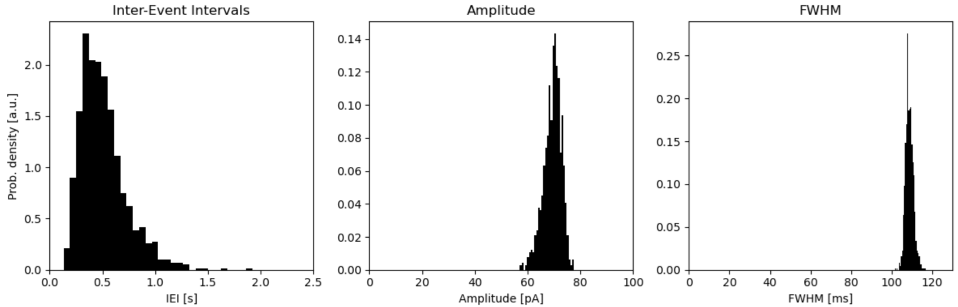


Results: Properties of Spontaneous SWRs

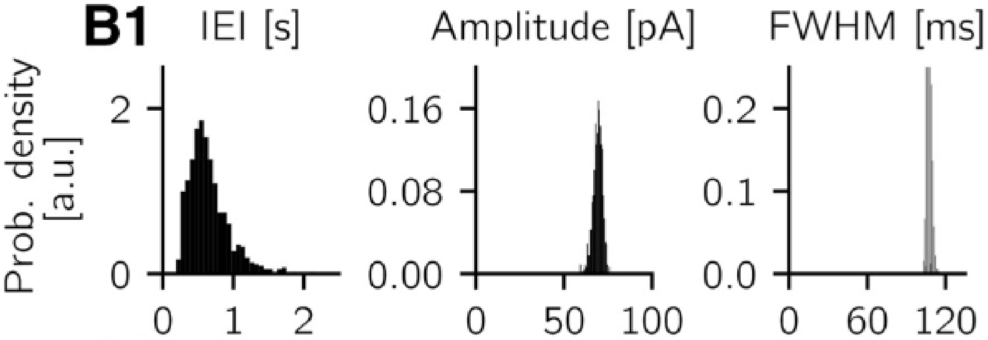
- SWR incidence: 1.0 - 1.55
- Average Event duration: 106.67



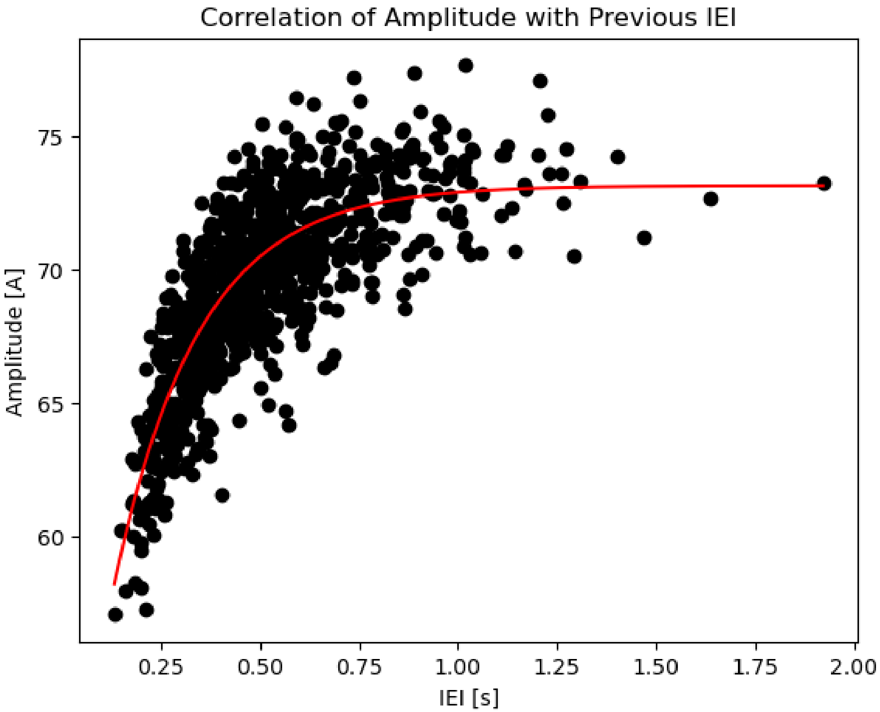
our results:



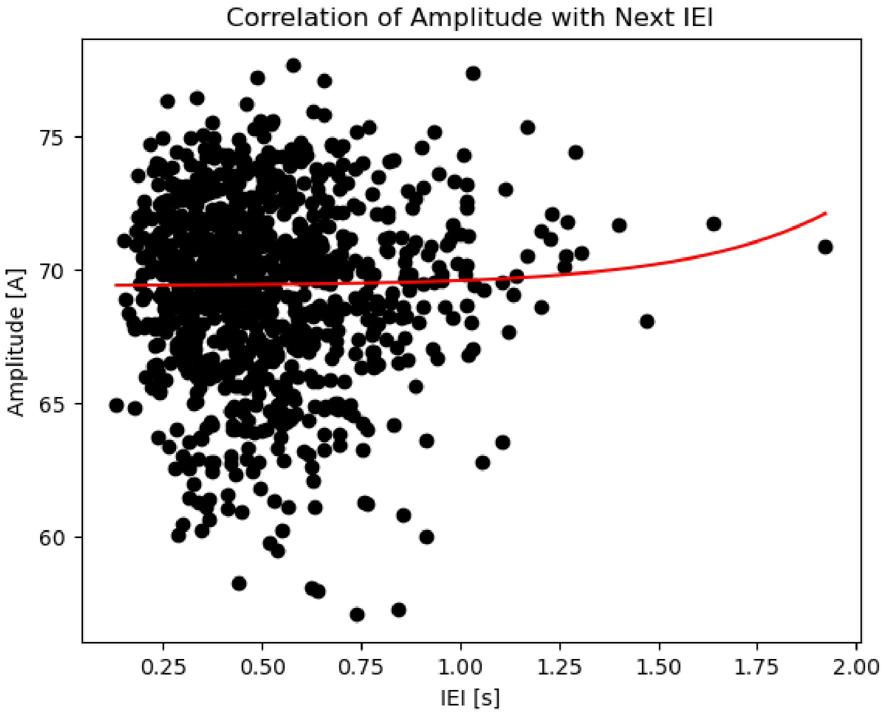
paper:



Results: Statistical Analysis



Pearson correlation coefficient: 0.696 with p-value: 0.000000000000



Pearson correlation coefficient: 0.004 with p-value: 0.889254867622

Summary

- could replicate several key findings of the paper
  - bistability
  - correlation of amplitude and previous IEI
    - in agreement with experimental results
  - similar range for values like incidence, FWHM

- promising mechanism to explain SWR emergence
  - but no neuron group(s) corresponding to A identified so far

References

- Evangelista, Roberta, et al. "Generation of sharp wave-ripple events by disinhibition." *Journal of Neuroscience* 40.41 (2020): 7811-7836.
- Buzsáki, G. (2015). Hippocampal sharp wave-ripple: A cognitive biomarker for episodic memory and planning. Hippocampus, 25(10), 1073-1188.
- https://www.researchgate.net/figure/The-hippocampus-is-the-ragion-where-memory-resides-Main-areas-are-CA1-CA3-and-DG_fig14_289479733
- https://sites.lsa.umich.edu/diba-lab/sharp-wave-ripples/

Supplementary Material - Estimation of LFP

- Estimation of LFP from synaptic currents to make findings comparable with electrophysiological recordings

Estimation procedure:

1. Record synaptic currents
2. Band-pass (90 Hz - 180 Hz) and Low-pass (< 5 Hz) filter the negative average signal
3. Apply peak detection script to Low-pass signal (Duarte, 2014)
4. Find onset of SWR event using FWHM according to Evangelista et al. (2022) and calculate event start and end

Supplementary Material - Estimation of LFP

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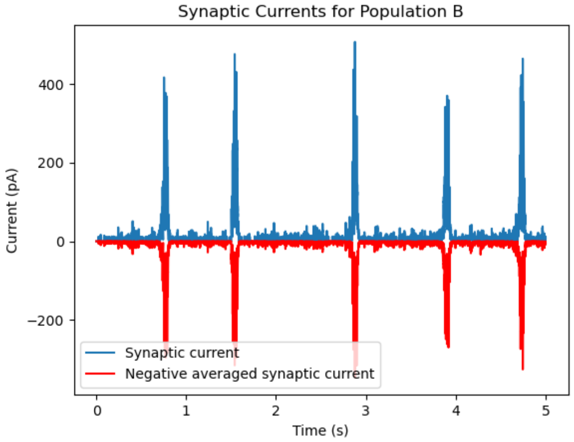

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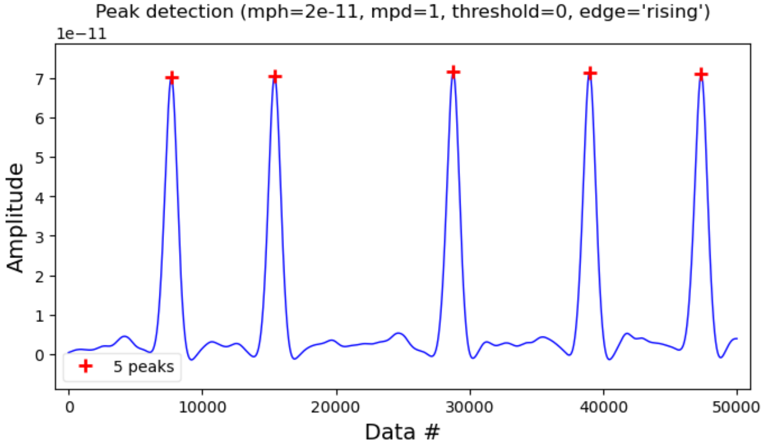

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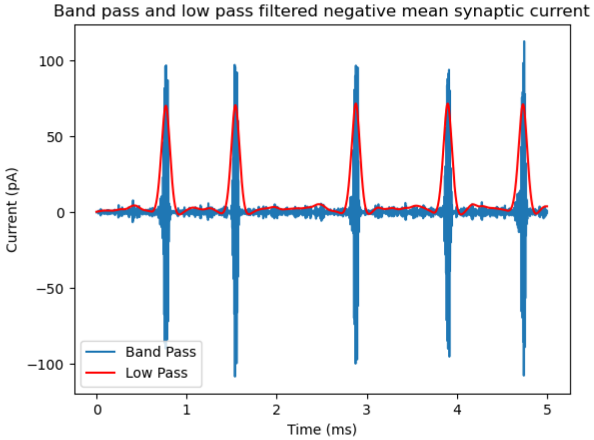

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